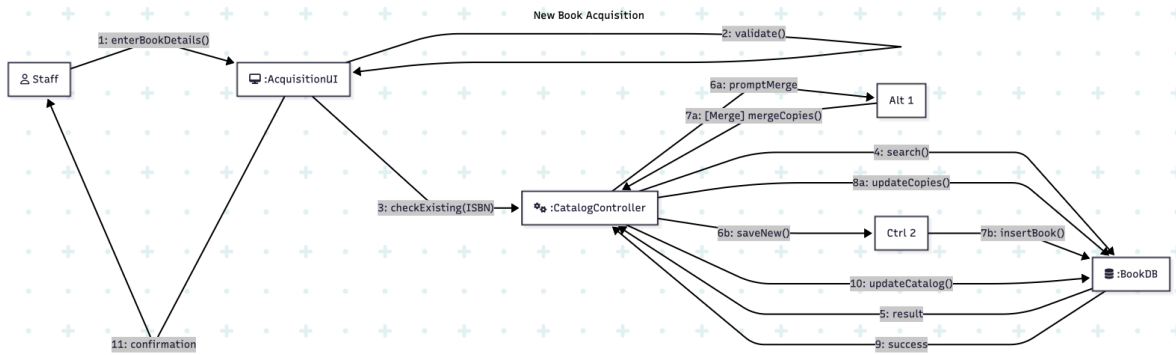
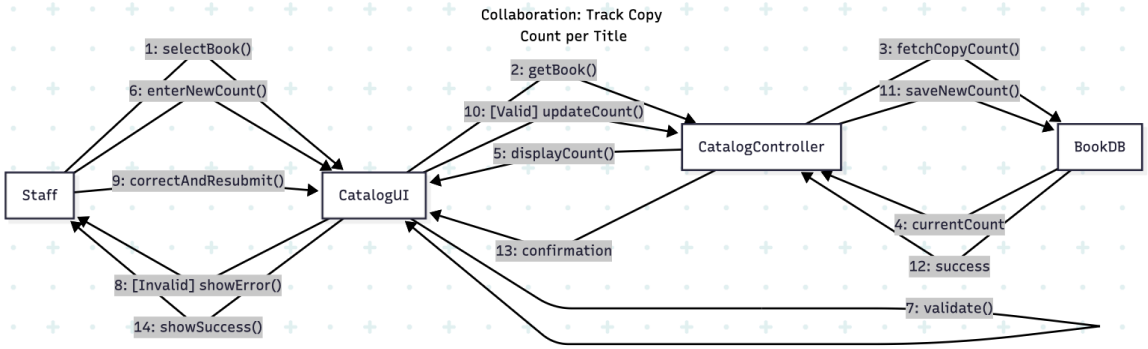
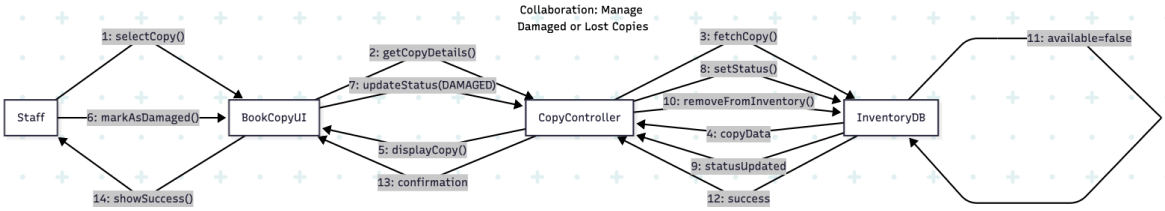
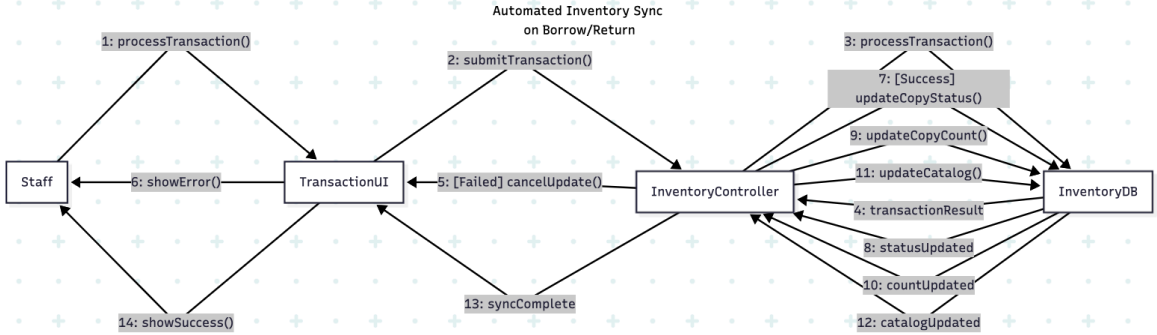
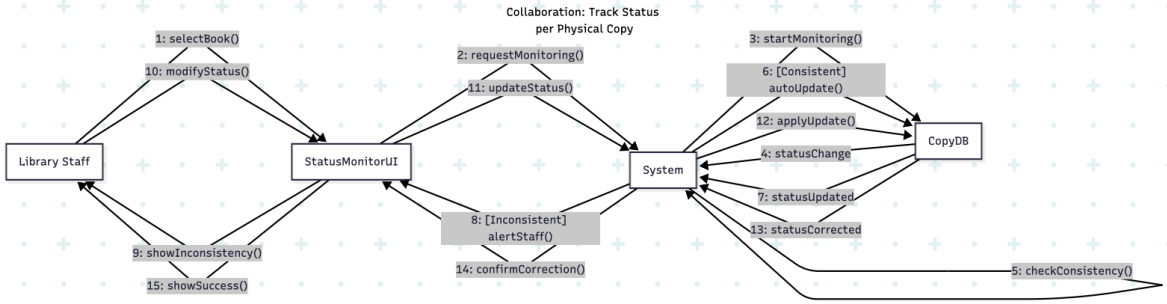
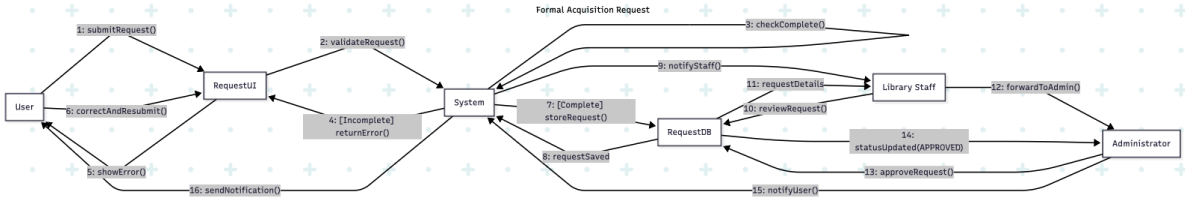

New Book Acquisition
Staff
1: enterBookDetails()
:AcquisitionUI
2: validate()
6a: promptMerge
Alt 1
7a: [Merge] mergeCopies()
4: search()
8a: updateCopies()
3: checkExisting(ISBN)
:CatalogController
6b: saveNew()
Ctrl 2
7b: insertBook()
:BookDB
10: updateCatalog()
5: result
9: success
11: confirmation

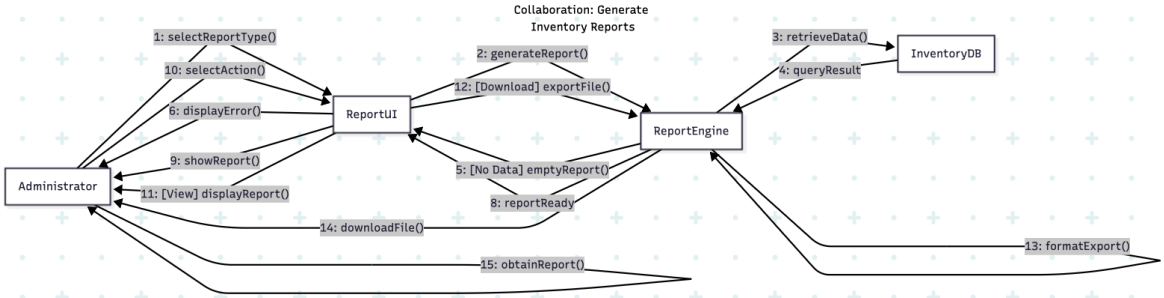

Collaboration: Track Copy Count per Title
1: selectBook()
6: enterNewCount()
2: getBook()
10: [Valid] updateCount()
5: displayCount()
3: fetchCopyCount()
11: saveNewCount()
Staff
9: correctAndResubmit()
CatalogUI
CatalogController
BookDB
4: currentCount
13: confirmation
12: success
8: [Invalid] showError()
14: showSuccess()
7: validate()


Collaboration: Manage
Damaged or Lost Copies
Staff
BookCopyUI
CopyController
InventoryDB
1: selectCopy()
2: getCopyDetails()
3: fetchCopy()
4: copyData
5: displayCopy()
6: markAsDamaged()
7: updateStatus(DAMAGED)
8: setStatus()
9: statusUpdated
10: removeFromInventory()
11: available=false
12: success
13: confirmation
14: showSuccess()


Automated Inventory Sync
on Borrow/Return
Staff
TransactionUI
InventoryController
InventoryDB
1: processTransaction()
2: submitTransaction()
3: processTransaction()
4: transactionResult
5: [Failed] cancelUpdate()
6: showError()
7: [Success]
updateCopyStatus()
8: statusUpdated
9: updateCopyCount()
10: countUpdated
11: updateCatalog()
12: catalogUpdated
13: syncComplete
14: showSuccess()


Collaboration: Track Status
per Physical Copy
Library Staff
StatusMonitorUI
System
CopyDB
1: selectBook()
2: requestMonitoring()
3: startMonitoring()
4: statusChange
5: checkConsistency()
6: [Consistent]
autoUpdate()
7: statusUpdated
8: [Inconsistent]
alertStaff()
9: showInconsistency()
10: modifyStatus()
11: updateStatus()
12: applyUpdate()
13: statusCorrected
14: confirmCorrection()
15: showSuccess()


Formal Acquisition Request
1: submitRequest()
2: validateRequest()
3: checkComplete()
9: notifyStaff()
11: requestDetails
12: forwardToAdmin()
User
RequestUI
System
Library Staff
6: correctAndResubmit()
4: [Incomplete] returnError()
7: [Complete] storeRequest()
10: reviewRequest()
RequestDB
14: statusUpdated(APPROVED)
Administrator
8: requestSaved
5: showError()
13: approveRequest()
16: sendNotification()
15: notifyUser()


Collaboration: Generate Inventory Reports
1: selectReportType()
3: retrieveData()
InventoryDB
2: generateReport()
10: selectAction()
4: queryResult
12: [Download] exportFile()
6: displayError()
ReportUI
ReportEngine
9: showReport()
5: [No Data] emptyReport()
Administrator
11: [View] displayReport()
8: reportReady
14: downloadFile()
13: formatExport()
15: obtainReport()